

Smart City Operations Network

Introduction

You have been assigned to design and implement the network infrastructure for Metropolis Smart City Operations, a centralized platform that connects all critical city services under a unified, fault-tolerant network. The system integrates real-time data from traffic management, public utilities, emergency dispatch, civic portals, and environmental monitoring stations spread across the city.

The network consists of six operational units:

Unit	Abbrevia tion	Devices	Notes
City Operations Hub	COH	500	Central backbone unit
Traffic Management Center	TMC	300	Manages all traffic signals and sensors
Public Utilities Node	PUN	220	Water, power, and waste monitoring
Emergency Dispatch Unit	EDU	180	Fire, police, ambulance coordination
Environmental Monitoring Post	EMP	90	Air quality, weather, flood sensors
Remote Field Station	RFS	50	Outpost with limited connectivity

Requirements

Topology Design & Server Setup

- Each unit is represented by a router. Each router must have at least two PCs and one printer connected to it via a switch.
- COH is the central backbone and connects directly to TMC, EDU, and EMP.
- TMC connects to PUN and EMP via a shared core switch (backbone switch network).
- PUN connects to EDU.
- RFS connects only to EMP. RFS has no other path to the rest of the network.
- COH hosts the central DNS server and a web server reachable at www.metropolis.city, displaying the message: "Smart City Operations - Online."
- Email servers must be configured at mail.coh.city and mail.tmc.city. Email communication must be configured between these two servers using both SMTP and POP3.
- Default routes are NOT allowed anywhere in the network.

Addressing

The base network address is derived from the student IDs of the first two members of your group using the following rule:

- Take the 3rd digit of Member 1's student ID. This becomes the first octet.
- Take the 5th digit of Member 2's student ID. This becomes the second octet.
- The third and fourth octets are 0, with a /16 prefix. The base network is therefore X.Y.0.0/16.

Example: Member 1 ID = 20231847 → 3rd digit = 2. Member 2 ID = 20204531 → 5th digit = 4. Base network = 2.4.0.0/16.

Apply VLSM subnetting in the following order: COH, TMC, PUN, EDU, EMP, RFS, then all point-to-point links last. Show your full VLSM tree as part of your submission.

DHCP Configuration

- COH uses a dedicated DHCP server to serve its own subnet. The COH router must forward DHCP requests to this server using an IP helper address (DHCP relay).
- TMC and PUN routers act as DHCP servers themselves, allocating IP addresses dynamically to their local devices.
- EDU and EMP also use their routers as DHCP servers for local devices.
- RFS uses static IP addressing only. DHCP must not be configured or used at RFS under any condition.
- All servers, routers, and printers across the entire network use static IP addresses.

DHCP pool starting addresses are derived from student IDs as follows:

- For the TMC DHCP pool: the starting host address of the pool begins at the offset equal to the last two digits of Member 1's student ID within the TMC subnet.
- For the PUN DHCP pool: the starting host address begins at the offset equal to the last two digits of Member 2's student ID within the PUN subnet.
- All other DHCP pools start from the first usable host address of their respective subnet.

Example: Member 1 ID = 20231847 → last two digits = 47. If the TMC subnet is 2.4.1.0/24, the DHCP pool starts at 2.4.1.47 instead of 2.4.1.1. Excluded addresses (2.4.1.1 through 2.4.1.46) must be configured accordingly.

Routing Design

All routing must be explicitly configured. Default routes are strictly prohibited. The following rules govern how each unit learns and advertises routes:

Static Routing

- Next-hop static routing: COH must use next-hop static routes to reach the EDU network directly.
- Exit-interface static routing: RFS must use exit-interface static routes to reach all other networks via EMP.
- Recursive static routing: EMP must use a recursive static route to reach the COH network, pointing to TMC as the intermediate hop. TMC must therefore already know the path to COH for this to resolve correctly.
- Floating static route: TMC must have a floating static route to the COH network via PUN as a backup path. Set the administrative distance of this route to 150, ensuring it only becomes active when the primary RIPv2-learned route to COH is unavailable.

Dynamic Routing (RIPv2)

- TMC, PUN, and EDU use RIPv2 to share routing information among themselves.
- COH participates in RIPv2 toward TMC and EMP, while also maintaining static routes toward EDU. COH must redistribute its static routes into the RIPv2 domain so that TMC and PUN learn about the EDU network dynamically.

Routing Constraints

- RFS must not participate in any dynamic routing protocol and can only be reached through EMP.
- PUN must not learn routes to COH directly from EDU - it must learn them only via RIPv2 through TMC.
- If the direct TMC-COH link goes down, TMC's floating static route via PUN must activate automatically, ensuring continuous reachability to COH.
- EMP must always be reachable from COH through at least two distinct paths. If the direct COH-EMP link fails, traffic must reroute through TMC using RIPv2 convergence.
- Routing must prevent loops at all times. No default routes are permitted anywhere in the network.
- All devices across all units must be able to ping one another after full configuration.

Deliverables

1. Cisco Packet Tracer .pkt / .pka file with complete, working configuration.
2. Network topology diagram with all routers, switches, servers, and point-to-point links labeled, showing subnet addresses as notes.
3. Full VLSM tree.
4. IP addressing table (unit, device, IP address, subnet mask, default gateway).
5. Full router configuration commands for all six routers.
6. Server configuration screenshots or commands (DNS, web, DHCP, email).
7. Work distribution table showing which group member configured which part.